MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROGRAM CONTROL UNIT, KEY CODE MJC

This amendment forms a part of MIL-M-28787/308A(EC), dated 23 July 1981, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 4.4.2, delete in its entirety and substitute:

"4.4.2 Order of testing. Test conditions A through LL shall be performed in any order with the following exceptions. Test conditions A, B, C, D, and E are stress tests and will be performed only at 25°C ±5°. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence."

PAGES 4, 5, 6, 7, AND 8

TABLE I: Delete and substitute new table I as printed on pages 2, 3, 4, 5, and 6 of this amendment.

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FIGURE 9: Delete and substitute new figure 9 as printed on page 7 of this amendment.

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FIGURE 12: Delete and substitute new figure 12 as printed on page 8 of this amendment.

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* FIGURE 31, NOTES: Add new note as follows: "3. Setups are performed by first connecting points A and B, then momentarily connecting point C to pin 4."

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* FIGURE 33, NOTES: Delete note 3 in its entirety.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	Voltage (V_{CC}) stress	III	Q	← Must pass all remaining tests →										
C1-C16	Input breakdown current (I_{IB})	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D16	Input clamp diode voltage	V	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E10	Output short circuit current (I_{OS})	VI	X	-30	-85	-30	-85	-30	-85	-30	-85	-30	-85	mA
F1-F4	Input current, high level	VII	X	0	20	0	20	0	20	0	20	0	20	µA
F5-F10,F12, F14, F15,	Input current, high level	VII	X	0	40	0	40	0	40	0	40	0	40	µA
F11	Input current, high level	VII	X	0	90	0	90	0	90	0	90	0	90	µA
F13	Input current, high level	VII	X	0	50	0	50	0	50	0	50	0	50	µA
F16	Input current, high level	VII	X	0	250	0	250	0	250	0	250	0	250	µA
G1-G4	Input current, low level	VIII	X	0.000	-0.398	0.000	-0.398	0.000	-0.398	0.000	-0.398	0.000	-0.398	mA
G5-G10,G12, G13, G15	Input current, low level	VIII	X	0.000	-0.775	0.000	-0.775	0.000	-0.775	0.000	-0.775	0.000	-0.775	mA
G11	Input current, low level	VIII	X	0.00	-2.55	0.00	-2.55	0.00	-2.55	0.00	-2.55	0.00	-2.55	mA
G14	Input current, low level	VIII	X	0.000	-0.725	0.000	-0.725	0.000	-0.725	0.000	-0.725	0.000	-0.725	mA
G16	Input current, low level	VIII	X	0.00	-3.59	0.00	-3.59	0.00	-3.59	0.00	-3.59	0.00	-3.59	mA
H1-H10	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J10	Output voltage, low level	X	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K4	High impedance leakage current, high level	XI	X	-1	+50	-1	+50	-1	+50	-1	+50	-1	+50	µA
L1-L4	High impedance leakage current, low level	XII	X	-50	+1	-50	+1	-50	+1	-50	+1	-50	+1	µA
M	Power supply current	XIII	X	10	208	10	218	10	218	10	239	10	231	mA
N1-N6	Functional tests	XIV	X	← See table XIV and notes →										
P1-P10	Transition time, t_{TLH}	XVI	S	2	17	2	17	2	21	2	17	2	25	ns
Q1-Q10	Transition time, t_{THL}	XVII	S	1	18	1	18	1	21	1	18	1	24	ns
R1	Propagation delay time, t_{PLH}	XVIII	S	7	70	7	70	7	79	7	70	8	88	ns
R2,R3	Propagation delay time, t_{PLH}	XVIII	S	5	58	5	58	6	66	5	58	7	74	ns
R4	Propagation delay time, t_{PLH}	XVIII	S	6	65	6	65	7	73	6	65	8	82	ns
R5	Propagation delay time, t_{PLH}	XVIII	S	7	78	7	78	8	87	7	78	9	97	ns
R6	Propagation delay time, t_{PLH}	XVIII	S	6	69	6	69	7	78	6	69	8	87	ns
R7	Propagation delay time, t_{PLH}	XVIII	S	5	53	5	53	6	60	5	53	6	68	ns
R8,R11	Propagation delay time, t_{PLH}	XVIII	S	3	39	3	39	4	45	3	39	5	52	ns
R9	Propagation delay time, t_{PLH}	XVIII	S	4	46	4	46	5	53	4	46	6	60	ns
R10	Propagation delay time, t_{PLH}	XVIII	S	6	61	6	61	6	69	6	61	7	78	ns
R12	Propagation delay time, t_{PLH}	XVIII	S	2	29	2	29	3	33	2	29	3	37	ns
R13	Propagation delay time, t_{PLH}	XVIII	S	2	21	2	21	2	25	2	21	3	30	ns
R14	Propagation delay time, t_{PLH}	XVIII	S	3	36	3	36	4	41	3	36	4	46	ns
R15,R16	Propagation delay time, t_{PLH}	XVIII	S	2	17	2	17	2	20	2	17	2	23	ns
R17	Propagation delay time, t_{PLH}	XVIII	S	5	59	5	59	6	66	5	59	7	74	ns
R18,R19	Propagation delay time, t_{PLH}	XVIII	S	4	46	4	46	5	52	4	46	5	58	ns
R20	Propagation delay time, t_{PLH}	XVIII	S	5	53	5	53	5	59	5	53	6	66	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R21	Propagation delay time, t_{PLH}	XVIII	S	6	66	6	66	7	75	6	66	8	84	ns
R22	Propagation delay time, t_{PLH}	XVIII	S	3	38	3	38	4	43	3	38	4	48	ns
R23,R24	Propagation delay time, t_{PLH}	XVIII	S	4	47	4	47	5	53	4	47	6	60	ns
R25	Propagation delay time, t_{PLH}	XVIII	S	4	43	4	43	4	49	4	43	5	55	ns
R26,R29	Propagation delay time, t_{PLH}	XVIII	S	2	28	2	28	3	33	2	28	3	38	ns
R27	Propagation delay time, t_{PLH}	XVII	S	3	35	3	35	4	40	3	35	4	45	ns
R28	Propagation delay time, t_{PLH}	XVIII	S	5	50	5	50	5	57	5	50	6	65	ns
S1	Propagation delay time, t_{PHL}	XIX	S	7	70	7	70	7	79	7	70	8	88	ns
S2,S3	Propagation delay time, t_{PHL}	XIX	S	5	58	5	58	6	66	5	58	7	74	ns
S4	Propagation delay time, t_{PHL}	XIX	S	6	65	5	65	7	73	6	65	8	82	ns
S5	Propagation delay time, t_{PHL}	XIX	S	7	78	7	78	8	87	7	78	9	97	ns
S6	Propagation delay time, t_{PHL}	XIX	S	6	69	6	69	7	78	6	69	8	87	ns
S7	Propagation delay time, t_{PHL}	XIX	S	5	53	5	53	6	60	5	53	6	68	ns
S8,S11	Propagation delay time, t_{PHL}	XIX	S	3	39	3	39	4	45	3	39	5	52	ns
S9	Propagation delay time, t_{PHL}	XIX	S	4	46	4	46	5	53	4	46	6	60	ns
S10	Propagation delay time, t_{PHL}	XIX	S	6	61	6	61	6	69	6	61	7	78	ns
S12	Propagation delay time, t_{PHL}	XIX	S	2	29	2	29	3	33	2	29	3	37	ns
S13	Propagation delay time, t_{PHL}	XIX	S	2	21	2	21	2	25	2	21	3	30	ns
S14	Propagation delay time, t_{PHL}	XIX	S	3	36	3	36	4	41	3	36	4	46	ns
S15,S16	Propagation delay time, t_{PHL}	XIX	S	2	17	2	17	2	20	2	17	2	23	ns
S18,S19	Propagation delay time, t_{PHL}	XIX	S	4	46	4	46	5	52	4	46	5	58	ns
S20	Propagation delay time, t_{PHL}	XIX	S	5	53	5	53	5	59	5	53	6	66	ns

See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S17	Propagation delay time, t_{PHL}	XIX	S	5	59	5	59	6	66	5	59	7	74	ns
S21	Propagation delay time, t_{PHL}	XIX	S	6	66	6	66	7	75	6	66	8	84	ns
S22	Propagation delay time, t_{PHL}	XIX	S	3	38	3	38	4	43	3	38	4	48	ns
S23,S24	Propagation delay time, t_{PHL}	XIX	S	4	47	4	47	5	53	4	47	6	60	ns
S25	Propagation delay time, t_{PHL}	XIX	S	4	43	4	43	4	49	4	43	5	55	ns
S26,S29	Propagation delay time, t_{PHL}	XIX	S	2	28	2	28	3	33	2	28	3	38	ns
S27	Propagation delay time, t_{PHL}	XIX	S	3	35	3	35	4	40	3	35	4	45	ns
S28	Propagation delay time, t_{PHL}	XIX	S	5	50	5	50	5	57	5	50	6	65	ns
T1-T4	Disable recovery time, t_{PLZ}	XX	S	5	55	5	55	6	60	5	55	6	65	ns
U1-U4	Enable recovery time, t_{PZL}	XXI	S	5	50	5	50	5	55	5	50	6	60	ns
V1-V4	Disable recovery time, t_{PHZ}	XXII	S	5	55	5	55	6	60	5	55	6	65	ns
W1-W4	Enable recovery time, t_{PZH}	XXIII	S	5	50	5	50	5	55	5	50	6	60	ns
X1-X9	Output crosstalk, low level	XXIV	Q	0	500	0	500	0	500	0	500	0	500	mV
Y1-Y8	Output crosstalk, high level	XXV	Q	0	500	0	500	0	500	0	500	0	500	mV
Z	Power supply transient amplitude	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
AA	Power supply filter capacitance	XXVII	S	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	3.15	5.85	µF
BB1-BB4	Input capacitance	XXVIII	Q	6	55	6	55	6	55	6	55	6	55	pF
BB5-BB11, BB13-BB16	Input capacitance	XXVIII	Q	2	25	2	25	2	25	2	25	2	25	pF

- - - See footnotes at end of table.

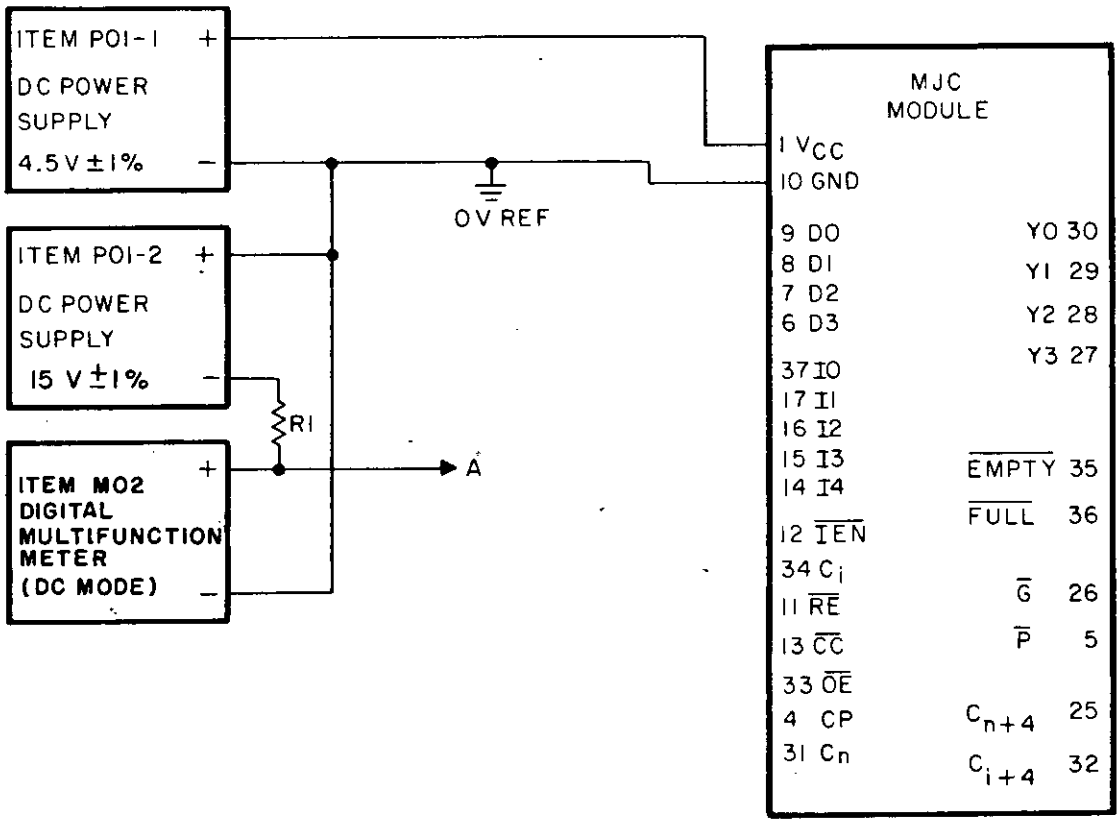
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Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
BB12	Input capacitance	XXVIII	Q	4	40	4	40	4	40	4	40	4	40	pF
CC1-CC10	Output capacitance	XXIX	Q	2	30	2	30	2	30	2	30	2	30	pF
DD1-DD4	Enable recovery time, t_{PZL}	XXX	S	8	80	8	80	10	100	8	80	11	106	ns
EE1-EE4	Disable recovery time, t_{PLZ}	XXXI	S	6	60	6	60	7	70	6	60	8	76	ns
FF1-FF4	Disable recovery time, t_{PHZ}	XXXII	S	6	60	6	60	7	70	6	60	8	76	ns
GG1-GG4	Enable recovery time, t_{PZH}	XXXIII	S	8	80	8	80	10	100	8	80	11	106	ns
HH1-HH4	Enable recovery time, t_{PZL}	XXXIV	S	6	60	6	60	7	70	6	60	8	76	ns
JJ1-JJ4	Disable recovery time, t_{PLZ}	XXXV	S	5	50	5	50	6	60	5	50	7	66	ns
KK1-KK4	Enable recovery time, t_{PZH}	XXXVI	S	6	60	6	60	7	70	6	60	8	76	ns
LL1-LL4	Disable recovery time, t_{PHZ}	XXXVII	S	5	50	5	50	6	60	5	50	7	66	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

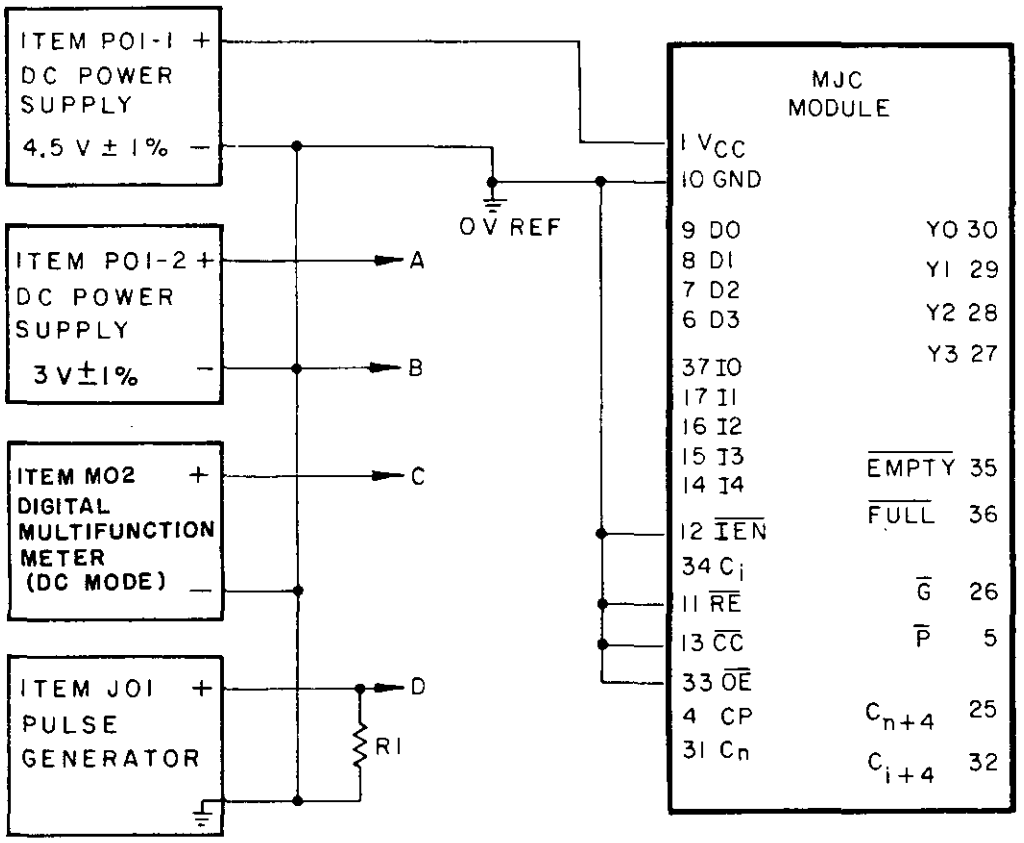
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NOTE: R1 = 750Ω ±1%, 1/4 W.

FIGURE 9. Input clamp diode voltage test circuit for table V.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load all outputs per figure 13.
3. Item J01 is used in the single pulse mode. When point D is referenced, apply a single pulse per figure 3.
4. For test condition J9, apply 17 single pulses after point D is referenced.

FIGURE 12. Output voltage (high-level and low-level) test circuit for tables IX and X.

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